Multimodal Hospitality Creator

Course Name: Gen AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: GAI-48

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MEDICAPS UNIVERSITY

Datagami
"Lead Digital Technology"

Problem Statement & Objectives

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I. Problem Statement & Objectives

1. **Problem Statement**
   Concept design in hospitality often suffers from a "vocabulary gap" between creative vision and technical execution. Designers struggle to instantly visualize complex architectural briefs, leading to slow iteration cycles and misaligned project expectations.

2. **Project Objectives**
   - To build an automated pipeline that bridges the gap between text-based narratives and high-fidelity visuals.
   - To implement a Retrieval-Augmented Generation (RAG) system to ensure AI outputs adhere to professional design standards.

3. **Scope of the Project**
   Focuses on four primary architectural styles (Biophilic, Desert Futurism, Heritage Revival, Alpine Minimalism) with the ability to generate both technical descriptions and photorealistic 8K imagery.

II. Proposed Solution

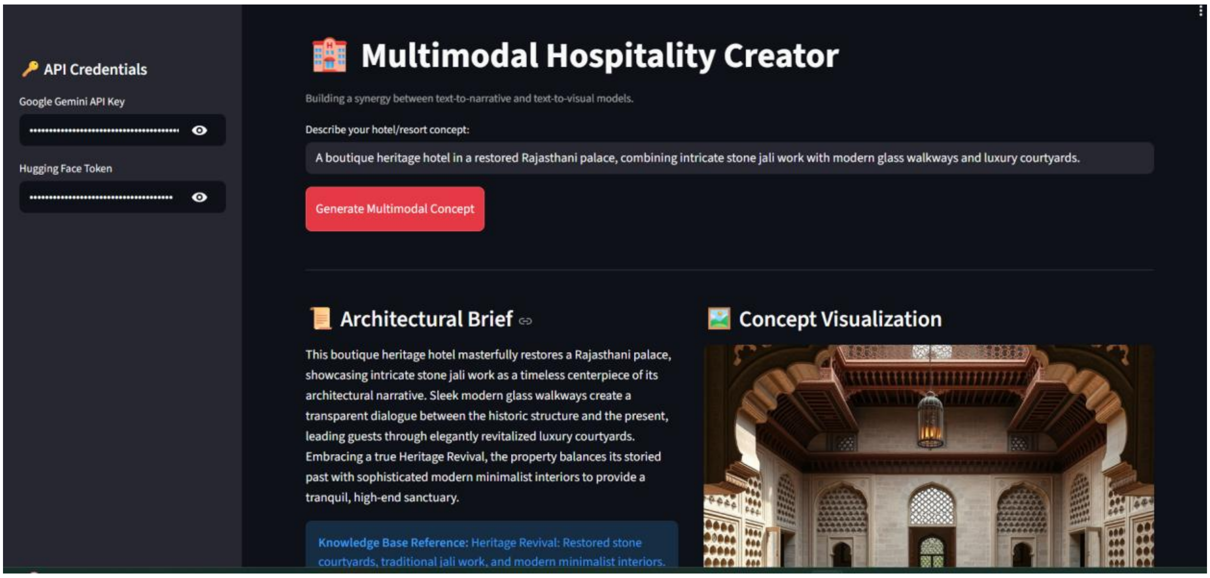
1. **Key Features**
   - Context-Aware Retrieval: Automatically pulls the most relevant architectural standards based on user input.
   - Multimodal Synthesis: Concurrent generation of structural narratives (text) and concept art (images).
   - Secure API Handling: Implementation of dynamic "Bring Your Own Key" (BYOK) architecture with input sanitization.

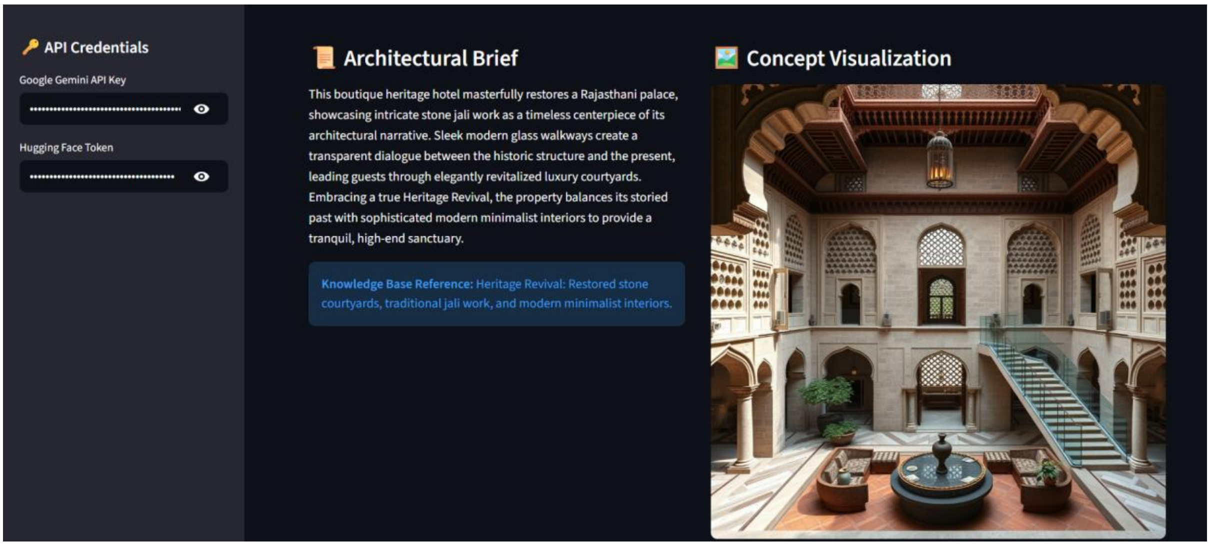
2. **Overall Architecture / Workflow**
   The application follows a Modular RAG Workflow:
   - Input Layer: User submits a vision via a Streamlit interface.
   - Retrieval Layer: ChromaDB performs a semantic search using Gemini Embeddings to find matching design guardrails.
   - Synthesis Layer: Gemini 3 Flash processes the prompt + retrieved data to write a brief, while the FLUX.1 model generates the visual concept.

3. Tools & Technologies Used

- Language: Python 3.12
- LLM: Google Gemini 3 Flash (Text) & FLUX.1 (Image)
- Vector DB: ChromaDB (for semantic memory)
- Orchestration: LangChain (connecting the models)
- UI: Streamlit

III. Results & Output





IV. Conclusion

The Multimodal Hospitality Creator proves that RAG is not just for chatbots—it is a powerful tool for professional creative industries. By grounding generative models in domain-specific data, we can create high-utility tools that move beyond generic AI and toward specialized, reliable software solutions.

V. Future Scope & Enhancements

- 3D Mesh Generation: Expanding the pipeline to generate 3D models from the 2D visual concepts.
- Cost Estimation Module: Using the generated brief to calculate an approximate "Cost Per Square Foot" based on material types (e.g., bamboo vs. stone).
- User Feedback Loop: Implementing a "Like/Dislike" feature that updates the ChromaDB index to refine the AI's understanding of "Luxury" over time.